# Applying DAX Fundamentals in Power BI: From Measures to Contextual Analysis

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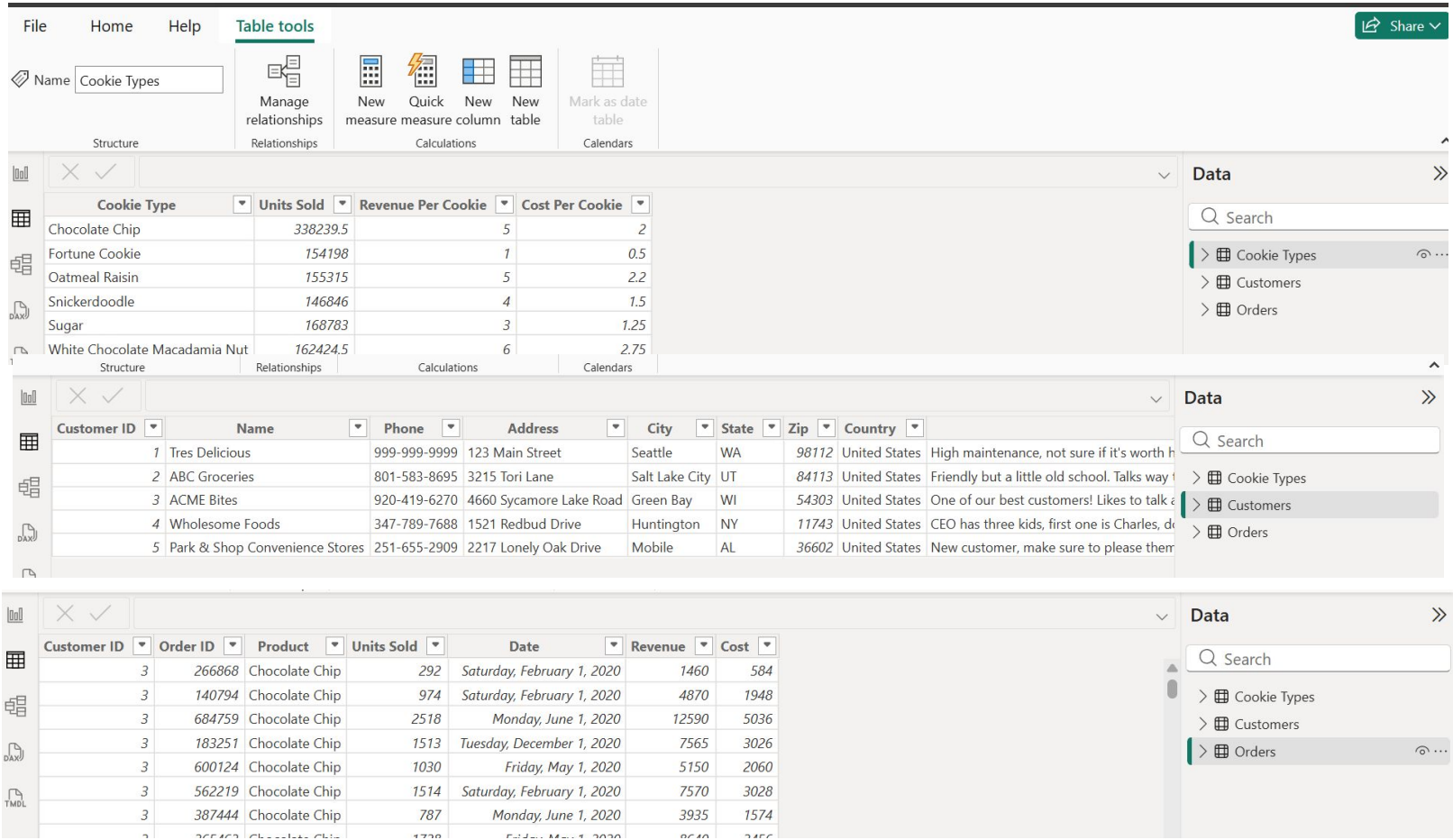
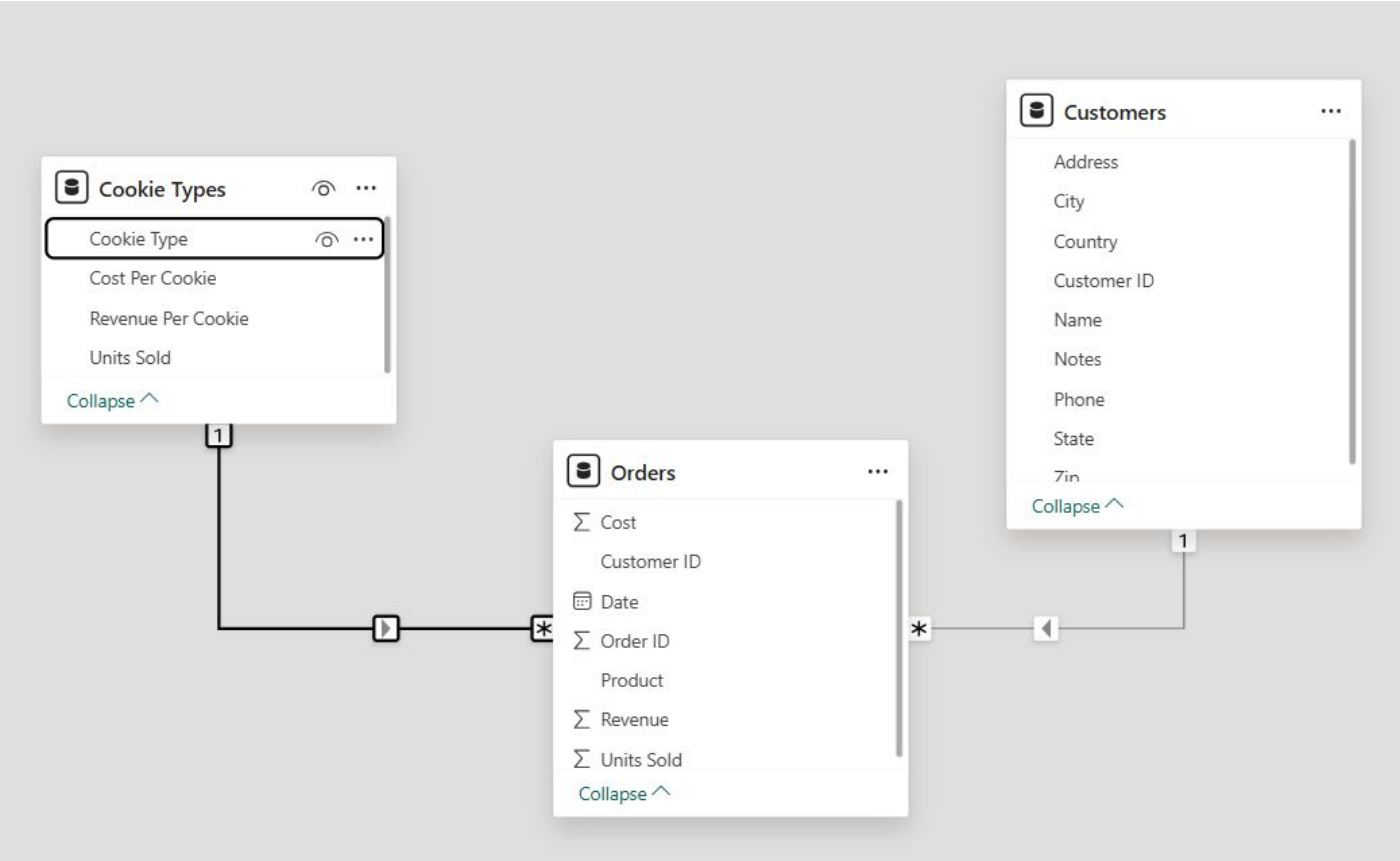
IS 107 - Business Intelligence

# Load the 3 data source

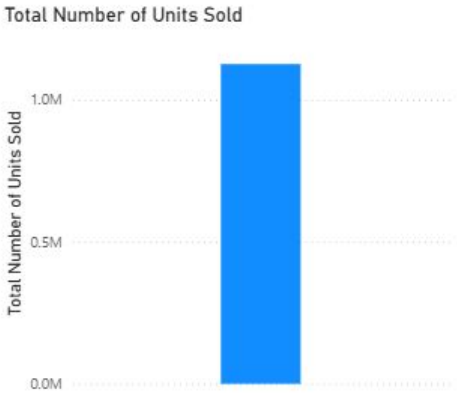


# Connecting Relationships of Cookie Type



# New Measure using Sum and the Units Sold

```
1 Total Number of Cookies Sold = SUM(Orders[Units Sold])
```




| Total Number of Units Sold |
|---|
| 1,125,806.00 |

| Name | Total Number of Unit Sold |
|---|---|
| ABC Groceries | 221,023 |
| ACME Bites | 329,677 |
| Park & Shop Convenience Stores | 179,448 |
| Tres Delicious | 128,769 |
| Wholesome Foods | 266,890 |
| **Total** | **1,125,806** |

# New Measure using COUNTROWS with the Orders

Structure | Formatting

```
1 Count fo Orders = COUNTROWS(Orders)
```

| Count fo Orders |
|---|
| 700 |

| Name | Count fo Orders |
|---|---|
| ABC Groceries | 132 |
| ACME Bites | 206 |
| Park & Shop Convenience Stores | 114 |
| Tres Delicious | 92 |
| Wholesome Foods | 156 |
| **Total** | **700** |

# New Measure using DISTINCTCOUNT with the Customers

| Distinct Customers |
|---|
| 5 |

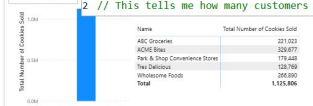
| Name | Distinct Customers |
|---|---|
| ABC Groceries | 1 |
| ACME Bites | 1 |
| Park & Shop Convenience Stores | 1 |
| Tres Delicious | 1 |
| Wholesome Foods | 1 |
| **Total** | **5** |

lame: Distinct Customers | Format: Whole number | Data category: Uncateg

lome table: Orders | 0

Structure | Formatting | Properties

```
1 Distinct Customers = DISTINCTCOUNT(Orders[Customer ID])
2 // This tells me how many customers we have
```



| Name | Total Number of Cookies Sold |
|---|---|
| ABC Groceries | 221,023 |
| ACME Bites | 329,677 |
| Park & Shop Convenience Stores | 179,448 |
| Tres Delicious | 128,789 |
| Wholesome Foods | 266,890 |
| **Total** | **1,125,806** |

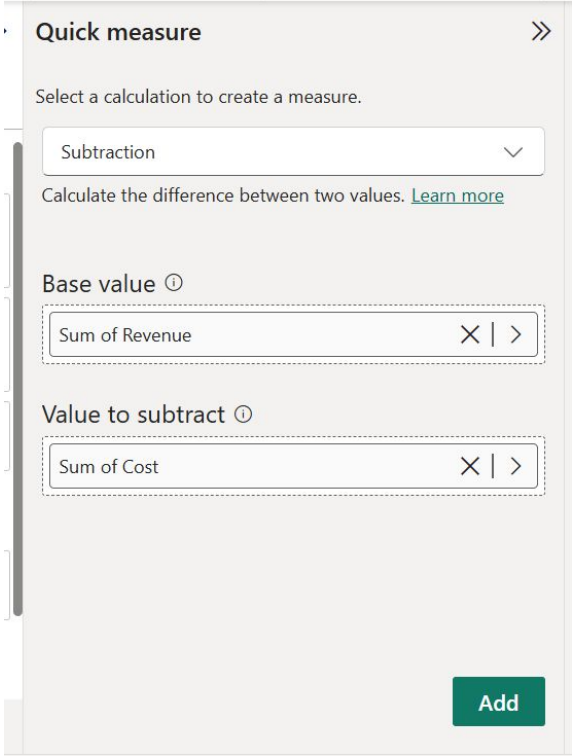
| Name | Count fo Orders |
|---|---|
| ABC Groceries | 132 |
| ACME Bites | 206 |
| Park & Shop Convenience Stores | 114 |
| Tres Delicious | 92 |
| Wholesome Foods | 156 |
| **Total** | **700** |

| Name | Distinct Customers |
|---|---|
| ABC Groceries | 1 |
| ACME Bites | 1 |
| Park & Shop Convenience Stores | 1 |
| Tres Delicious | 1 |
| Wholesome Foods | 1 |
| **Total** | **5** |

Using comments to remember what is the measure about.

# New Measure using Quick Measure Finding the Total Profit

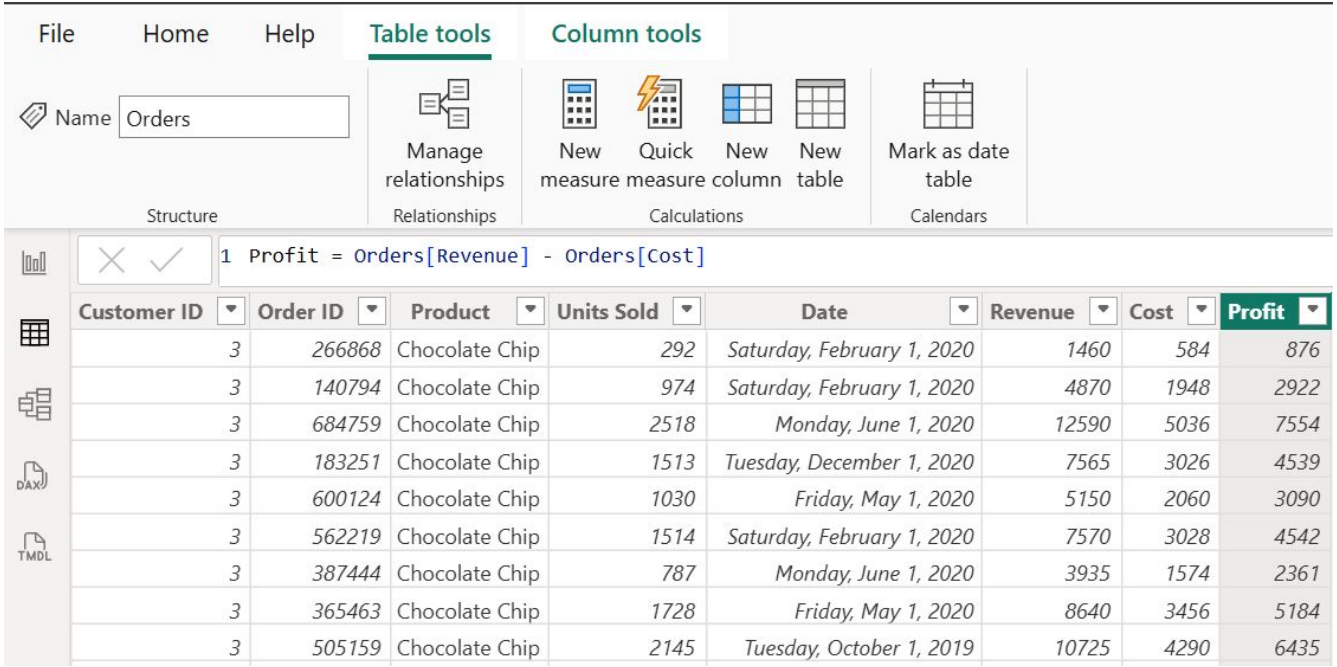
| Name | Total Profit |
|---|---|
| ABC Groceries | 523,317.00 |
| ACME Bites | 828,379.00 |
| Park & Shop Convenience Stores | 423,498.50 |
| Tres Delicious | 302,199.00 |
| Wholesome Foods | 639,677.00 |
| **Total** | **2,717,070.50** |




Using quick measure

| Name | Total Profit | Revenue minus Sum of Cost |
|---|---|---|
| ABC Groceries | 523,317.00 | 523,317.00 |
| ACME Bites | 828,379.00 | 828,379.00 |
| Park & Shop Convenience Stores | 423,498.50 | 423,498.50 |
| Tres Delicious | 302,199.00 | 302,199.00 |
| Wholesome Foods | 639,677.00 | 639,677.00 |
| **Total** | **2,717,070.50** | **2,717,070.50** |

# New Column by adding column in table using the New Column




| Customer ID | Order ID | Product | Units Sold | Date | Revenue | Cost | Profit |
|---|---|---|---|---|---|---|---|
| 3 | 266868 | Chocolate Chip | 292 | Saturday, February 1, 2020 | 1460 | 584 | 876 |
| 3 | 140794 | Chocolate Chip | 974 | Saturday, February 1, 2020 | 4870 | 1948 | 2922 |
| 3 | 684759 | Chocolate Chip | 2518 | Monday, June 1, 2020 | 12590 | 5036 | 7554 |
| 3 | 183251 | Chocolate Chip | 1513 | Tuesday, December 1, 2020 | 7565 | 3026 | 4539 |
| 3 | 600124 | Chocolate Chip | 1030 | Friday, May 1, 2020 | 5150 | 2060 | 3090 |
| 3 | 562219 | Chocolate Chip | 1514 | Saturday, February 1, 2020 | 7570 | 3028 | 4542 |
| 3 | 387444 | Chocolate Chip | 787 | Monday, June 1, 2020 | 3935 | 1574 | 2361 |
| 3 | 365463 | Chocolate Chip | 1728 | Friday, May 1, 2020 | 8640 | 3456 | 5184 |
| 3 | 505159 | Chocolate Chip | 2145 | Tuesday, October 1, 2019 | 10725 | 4290 | 6435 |

Using quick measure

# New Measure for Profit Margin using Division

```
1 Profit Margin % = [Total Profit] / SUM(Orders[Revenue])
```

| Cookie Type | Profit Margin % |
|---|---|
| Chocolate Chip | 60.00% |
| Fortune Cookie | 50.00% |
| Oatmeal Raisin | 56.00% |
| Snickerdoodle | 62.50% |
| Sugar | 58.33% |
| White Chocolate Macadamia Nut | 54.17% |
| **Total** | **57.93%** |

# New Measure for Total Profit using Iterator Function

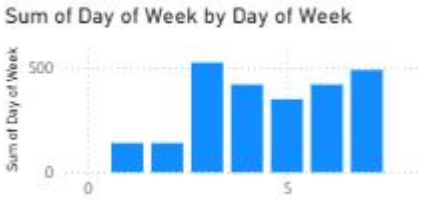
```
Total Profit 2 = SUMX('Cookie Types', 'Cookie Types'[Units Sold] * ('Cookie Types'[Revenue Per Cookie] - 'Cookie Types'[Cost Per Cookie]))
```

| Total Profit 2 |
|---|
| 2,717,064.38 |

# New Column using DATE Function

```
Day of Week = WEEKDAY(Orders[Date], 1)
```

| Customer ID | Order ID | Product | Units Sold | Date | Revenue | Cost | Profit | Day of Week |
|---|---|---|---|---|---|---|---|---|
| 3 | 266868 | Chocolate Chip | 292 | Saturday, February 1, 2020 | 1460 | 584 | 876 | 7 |
| 3 | 140794 | Chocolate Chip | 974 | Saturday, February 1, 2020 | 4870 | 1948 | 2922 | 7 |
| 3 | 684759 | Chocolate Chip | 2518 | Monday, June 1, 2020 | 12590 | 5036 | 7554 | 2 |
| 3 | 183251 | Chocolate Chip | 1513 | Tuesday, December 1, 2020 | 7565 | 3026 | 4539 | 3 |
| 3 | 600124 | Chocolate Chip | 1030 | Friday, May 1, 2020 | 5150 | 2060 | 3090 | 6 |



# New Column to find products with chocolate using FIND Function

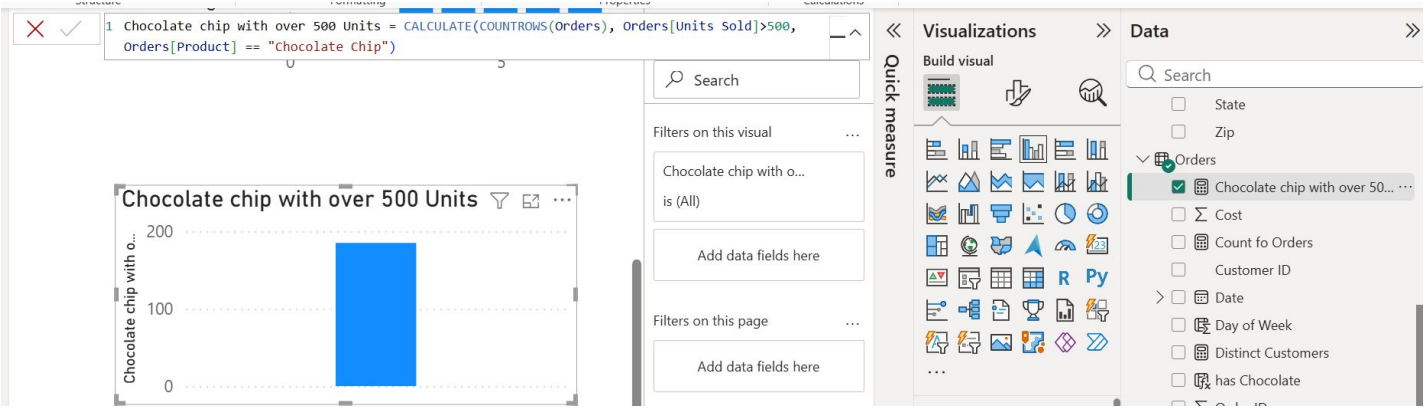
```
has Chocolate = FIND("Chocolate", Orders[Product],1,0)
```

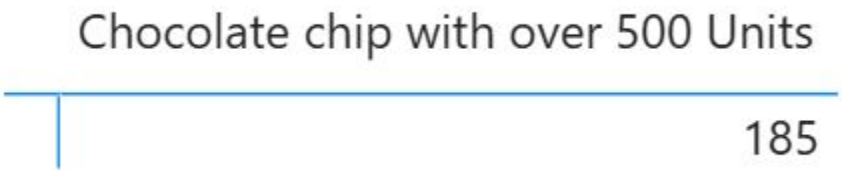
| Customer ID | Order ID | Product | Units Sold | Date | Revenue | Cost | Profit | Day of Week | has Chocolate |
|---|---|---|---|---|---|---|---|---|---|
| 3 | 266868 | Chocolate Chip | 292 | Saturday, February 1, 2020 | 1460 | 584 | 876 | 7 | 1 |
| 3 | 140794 | Chocolate Chip | 974 | Saturday, February 1, 2020 | 4870 | 1948 | 2922 | 7 | 1 |
| 3 | 684759 | Chocolate Chip | 2518 | Monday, June 1, 2020 | 12590 | 5036 | 7554 | 2 | 1 |

```
has Chocolate = IF(FIND("Chocolate", Orders[Product],1,0)>0,"Has Chocolate", "No Chocolate")
```

| Customer ID | Order ID | Product | Units Sold | Date | Revenue | Cost | Profit | Day of Week | has Chocolate |
|---|---|---|---|---|---|---|---|---|---|
| 3 | 266868 | Chocolate Chip | 292 | Saturday, February 1, 2020 | 1460 | 584 | 876 | 7 | Has Chocolate |
| 3 | 140794 | Chocolate Chip | 974 | Saturday, February 1, 2020 | 4870 | 1948 | 2922 | 7 | Has Chocolate |
| 3 | 684759 | Chocolate Chip | 2518 | Monday, June 1, 2020 | 12590 | 5036 | 7554 | 2 | Has Chocolate |
| 3 | 183251 | Chocolate Chip | 1513 | Tuesday, December 1, 2020 | 7565 | 3026 | 4539 | 3 | Has Chocolate |
| 3 | 600124 | Chocolate Chip | 1030 | Friday, May 1, 2020 | 5150 | 2060 | 3090 | 6 | Has Chocolate |
| 3 | 562219 | Chocolate Chip | 1514 | Saturday, February 1, 2020 | 7570 | 3028 | 4542 | 7 | Has Chocolate |
| 3 | 387444 | Chocolate Chip | 787 | Monday, June 1, 2020 | 3935 | 1574 | 2361 | 2 | Has Chocolate |
| 3 | 365463 | Chocolate Chip | 1728 | Friday, May 1, 2020 | 8640 | 3456 | 5184 | 6 | Has Chocolate |

# New Measure to find chocolate chip with over 500 units using Calculate Function



# GUIDE QUESTIONS:

## I. Foundational Concepts

1. Defining DAX: In your own words, how would you explain the difference between a standard Excel formula and a Data Analysis Expression (DAX) based on the tutorial?

**ANSWER:** Unlike Excel formulas that target specific cells, DAX operates across entire tables and relationships. It doesn't just crunch numbers, it understands *context*: who's filtering, where the data lives, and how tables connect. It is less of a calculator, more as a query engine into the data model.

2. The "Why" of DAX: Why can’t we just use the raw numbers from our imported dataset to create complex reports? What is the specific "gap" that DAX fills in Power BI?

**ANSWER:** Raw numbers don't respond to filters, slicers, or table relationships. DAX fills that gap by making calculations dynamic. DAX is when you need a total that updates when you select a region, or a margin that shifts per category. Without it, your report is just a typical table.

# GUIDE QUESTIONS:

## II. Calculated Columns vs. Measures

3. The Storage Difference: The video demonstrates creating both Calculated Columns and Measures. Which one actually increases the file size of your Power BI model, and why is this an important consideration for large datasets?

**ANSWER:** Calculated Columns are stored physically in the model,one value per row, so they add to file size. Measures calculate only and store nothing. This gap matters: thousands of stored column values bloat your file, while measures stay lightweight no matter how large your dataset grows.

4. Context Clues: * When you created a Calculated Column, the calculation happened row-by-row.
○ When you created a Measure, the result changed depending on which filters or visuals you clicked.
○ Question: Explain how "Filter Context" affects the result of a Measure in your final dashboard.

**ANSWER:** Filter context is simply what's currently selected or filtered. When a measure hits a visual, Power BI checks what's active — a year slicer, a product segment — and recalculates accordingly. The same measure can return different results depending on where it lives or what the user clicks. One formula, infinite contexts.

# GUIDE QUESTIONS:

## III. Function Application & Syntax

5. The RELATED Function: In the tutorial, the RELATED function is used. Why is this function necessary when working with multiple tables, and what must be established in the Model View for this function to work correctly?

**ANSWER:** RELATED pulls a value from another table, but only when a relationship exists between them in Model View. No relationship, no result (just an error).

6. Deconstructing the Formula: Look at the following formula from the video:
Total Sales = SUM(Sales[Sales Amount]) Identify the Measure Name, the Function, the Table, and the Column. Why is it "best practice" to include the table name in the reference?

**ANSWER:** Including the table name (`Sales[Sales Amount]` instead of just `[Sales Amount]`) is best practice because it avoids confusion, especially when multiple tables have columns with similar names. It also makes your formula easier to read and debug later.

# GUIDE QUESTIONS:

## IV. Analysis & Troubleshooting

7. Data Interpretation: Look at the visual you created using your DAX measures. If the "Total Profit" measure shows a negative number for a specific category, does that mean the DAX formula is "broken," or is it telling you something about the business data? How can you verify which it is?

**ANSWER:** A negative number doesn't mean the formula is broken — it likely reflects reality, about the business, like costs exceeding revenue for that category. To verify, cross-check the measure against the raw data for that specific category. If your formula logic is right (e.g., `Profit = Revenue - Cost`) and costs are genuinely higher, the measure is working fine. It's just being honest.

8. Optimization: If you were tasked with calculating Profit Margin Percentage, would it be more efficient to create a Calculated Column for every row or a single Measure? Justify your answer based on what you learned about DAX efficiency.

**ANSWER:** Use a Measure, not a Calculated Column. Profit Margin % needs to react to filters and visual context while a column calculates row by row and can't aggregate properly across categories or time periods. A measure stays dynamic, lightweight, and accurate. Crucially, it gives you the margin of the filtered total, not an average of row-level margins.